

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
CO2 ASSESSMENT TOOL 2 – Intelligent Nuclear Power Plant Monitoring Platform

Course Code:	CSA10 / CSA1007	Course Title:	Software Engineering
CO Assessed:	CO2	Assessment:	Assessment Tool 2 :Intelligent Nuclear Power Plant Monitoring Platform
Weightage:	20%	Total Marks:	25
CO2 Statement:	<i>Perform detailed requirements analysis, design scalable architectures, and prioritize features with a focus on cross-disciplinary skills</i>		
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Date:	30-07-2026	Duration:	60 minutes

Code of Conduct

I _____ V.Naga Revanth _____ /192411128_____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.



Signature of the Student: _____

Annexure A – Intelligent Nuclear Power Plant Monitoring Platform

1. System Requirement Analysis

1.1 System Objectives

The Intelligent Nuclear Power Plant Monitoring Platform is designed to continuously monitor the operational status of a nuclear power plant using IoT sensors, Artificial Intelligence (AI), and cloud technologies. The system detects abnormal conditions, predicts equipment failures, provides real-time alerts, and supports decision-making to improve plant safety, efficiency, and reliability.

Objectives

- Monitor reactor temperature, pressure, radiation, and coolant levels.
- Detect abnormal operating conditions.
- Predict equipment failures using AI.
- Generate instant alerts for emergencies.
- Store monitoring data securely.
- Provide dashboards for operators and administrators.
- Improve operational safety and reduce downtime.

1.2 Functional Requirements

User Management

- User registration and login
- Role-based access (Admin, Engineer, Operator)

Sensor Monitoring

- Collect temperature data
- Monitor radiation levels
- Monitor pressure
- Monitor coolant flow
- Collect vibration data

AI Prediction Module

- Analyze sensor data
- Predict equipment failure
- Detect anomalies

Alert Management

- SMS notifications
- Email notifications
- Dashboard alerts
- Emergency shutdown notification

Dashboard

- Live sensor values
- Graphs
- Historical reports
- Equipment status

Data Management

- Store sensor history
- Backup data
- Generate reports

1.3 Non-Functional Requirements

Performance

- Real-time monitoring
- Response time below 2 seconds

Security

- Data encryption

- Multi-factor authentication
- Secure API communication

Reliability

- 24×7 monitoring
- Automatic fault recovery

Scalability

- Support thousands of sensors
- Easy expansion for multiple plants

Availability

- 99.99% uptime

Maintainability

- Modular architecture
- Easy software updates

1.4 Quality Attributes

Attribute	Description
Scalability	Supports additional sensors and plants
Reliability	Continuous operation with minimal failures
Performance	Fast processing of sensor data
Availability	High uptime with backup services
Security	Secure communication and authentication
Maintainability	Easy modification and testing

2. Selection of Software Architecture

Selected Architecture

Hybrid Architecture

(Client–Server + Layered + Event-Driven + Microservices)

Why Hybrid Architecture?

A nuclear power plant requires:

- Continuous monitoring
- Real-time alerts

- High reliability
- AI prediction
- Secure communication
- Independent services

A Hybrid Architecture combines the advantages of multiple architectural styles.

Layered Architecture

- Separates presentation, business logic, and data layers.
- Easier maintenance.

Client–Server

- Operators access centralized monitoring servers.

Event-Driven

- Immediate response to abnormal sensor values.

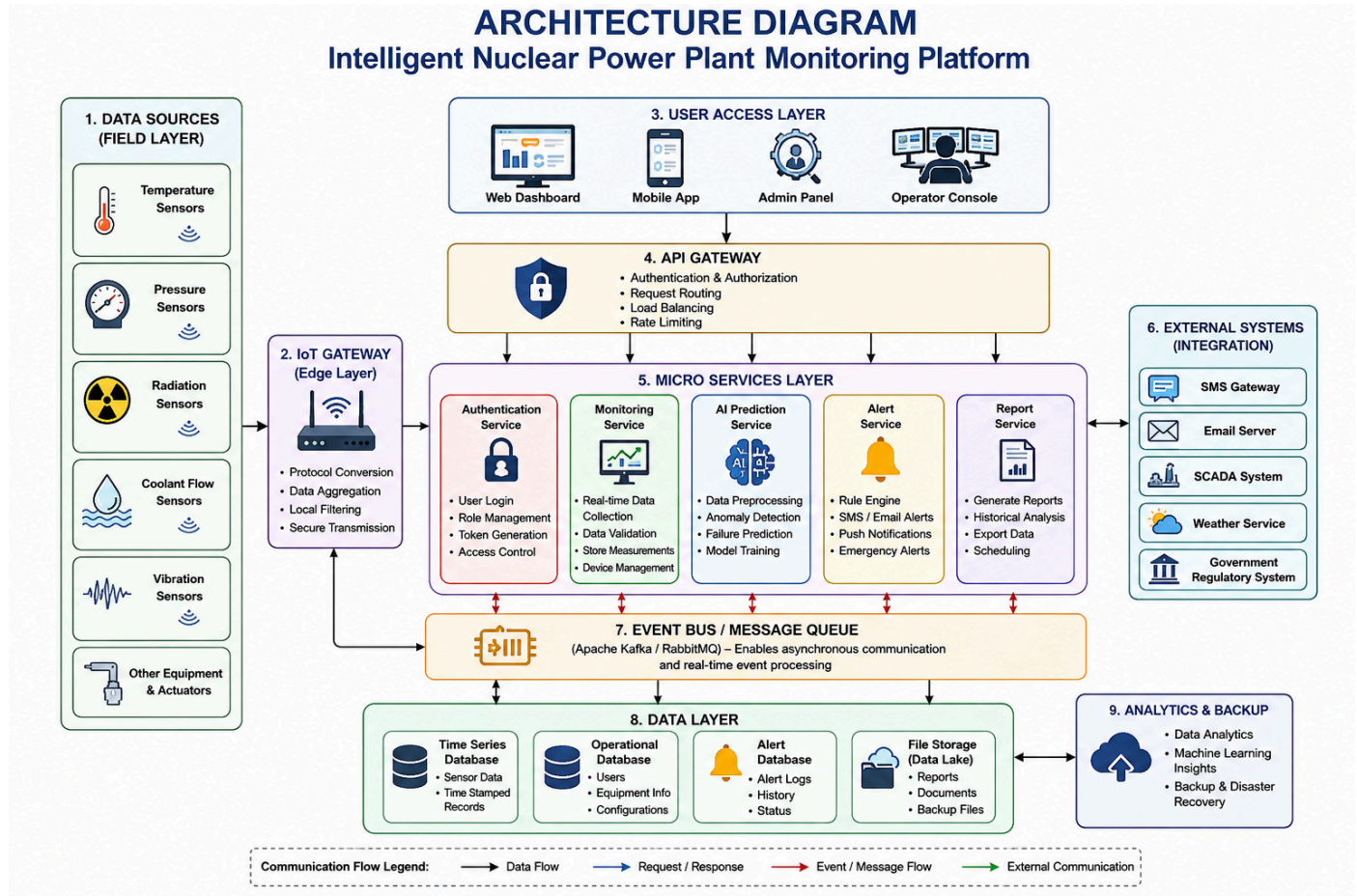
Microservices

- Independent AI service
- Notification service
- Authentication service

Advantages

- Highly scalable
- Secure
- Fault tolerant
- Easy maintenance
- Supports future expansion

3. Software Architecture Diagram



4. Components and Their Interactions

1. Web/Mobile Dashboard

Responsibilities:

- User login
- Display sensor values
- Show alerts
- Generate reports

2. API Gateway

Responsibilities:

- Receives requests

- Authentication
- Request routing
- Load balancing

3. Authentication Service

Responsibilities:

- Login
- User verification
- Role management
- Token generation

4. Monitoring Service

Responsibilities:

- Receive sensor data
- Validate data
- Store measurements

5. AI Prediction Service

Responsibilities:

- Analyze sensor history
- Detect anomalies
- Predict equipment failures

6. Alert Service

Responsibilities:

- Emergency alerts
- SMS
- Email
- Push notifications

7. Report Service

Responsibilities:

- Daily reports

- Monthly reports
- Historical analysis

8. Sensor Processing Module

Responsibilities:

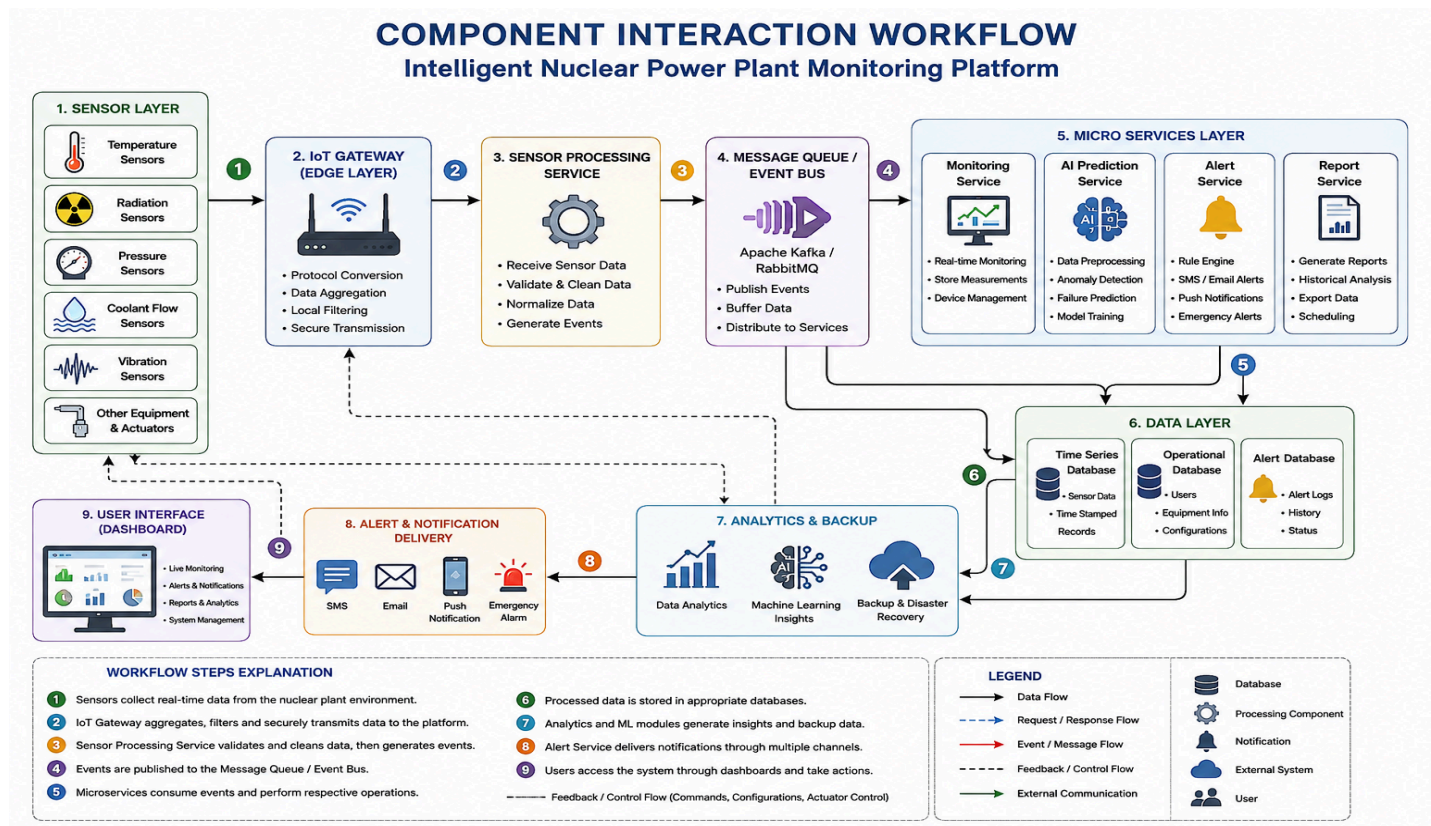
- Collect IoT sensor readings
- Clean sensor data
- Send events

9. Cloud Database

Responsibilities:

- Store sensor history
- User data
- Alert history
- Reports

Component Interaction Workflow



5. Quality Attribute Analysis

Scalability

- Microservices can scale independently.
- Supports additional sensors without affecting existing services.
- Cloud deployment enables horizontal scaling.

Security

- HTTPS communication
- JWT authentication
- Role-based access control
- Encrypted database
- Firewall protection

Performance

- Event-driven processing reduces latency.
- Message queue enables asynchronous communication.
- AI service processes data independently.

Reliability

- Backup database
- Fault-tolerant microservices
- Automatic recovery
- Redundant servers

Maintainability

- Modular services
- Independent deployment
- Easier debugging and testing
- Reusable components

Availability

- Cloud hosting
- Load balancing

- Database replication z.
- 24×7 monitoring

6. Architectural Justification

Reasons for Choosing Hybrid Architecture

- Supports real-time sensor monitoring.
- Allows AI modules to operate independently.
- Ensures continuous monitoring with minimal downtime.
- Enables secure communication between sensors and users.
- Facilitates future integration with cloud platforms and additional nuclear plants.

Trade-offs

Advantage	Trade-off
High scalability	Increased deployment complexity
Independent services	More infrastructure required
Better fault isolation	Higher monitoring overhead
Easy maintenance	Requires service orchestration

Future Enhancements

- Integration with digital twin technology.
- Predictive maintenance using deep learning.
- Autonomous emergency response.
- Drone-based inspection.
- Edge AI for faster processing.
- Multi-plant centralized monitoring.
- Blockchain for secure audit logs.

Conclusion

The proposed Hybrid Architecture (Layered + Client–Server + Event-Driven + Microservices) is well suited for the Intelligent Nuclear Power Plant Monitoring Platform because it provides high scalability, reliability, security, maintainability, and real-time performance. The architecture enables efficient monitoring of critical plant operations, AI-based anomaly detection, and rapid emergency response while supporting future expansion and long-term system sustainability.